a) 0.999×10^{099} $\quad \underbrace{0}_{s}\ \underbrace{099}_{exp.}\ 999$ $\quad 9.99 \times 10^{99}$

b) 00.10×10^{199} $\quad 0\ 199\ 010$ $\quad 0.10 \times \left(\frac{1}{10}\right)^{99}$

$\frac{1}{10} \leq m$

c) 10.10×10^{199} $\quad (-1) \cdot 0.10 \times \left(\frac{1}{10}\right)^{99}$

d) 7.8125×10^{-3} $\quad 7.81 \times 10^{-3}$ will be represented.

overflow

$E_t = 0.0025$ $\quad \varepsilon_t = \frac{2.5 \times 10^{-3}}{7.81 \times 10^{-3}}\% \cong 0.032\%$

e) We need two more bits in a word.

f) Overflow occurs when a number can't represented in the current word size.

g)